TSA Mock

Problem Solving

The hardest past TSA Problem Solving question of every type, from all 16 papers (2008–2023)

Questions	Total marks	Time limit
25	25	45 minutes

Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Every question is a real past TSA question. The year and number in brackets show where it comes from, e.g. **7 (2019 Q32)**.

What is in this paper

At least one question of each of the 19 Problem Solving question types that have come up in past papers, chosen as the hardest of its type: 2D shapes, tiling & rotations, 3D views & cubes, Algebra & unknowns, Area, perimeter & volume, Averages, Calendars, clocks & digital displays, Charts & graphs, Choosing a method, Counting & combinations, Logic & deduction, Maps, routes & networks, Money & best value, Nets & folding, Number patterns, cycles & codes, Percentages, Rates, ratio & proportion, Reading tables & data, Speed, distance & time, Time zones & timetables.

How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to 25 questions: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes). These are hard questions, so expect to run short of time.

Score: _____ / 25

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1 (2013 Q6) On only 360 occasions during the twentieth century was it possible to write the date in the form DD/MM/YYYY using eight different digits. One example was 28th July 1956 (28/07/1956).

Eleven of Arthur's birthdays have been such dates (though not his date of birth), and in 1974 and 1983 the two digits that did not form part of the date made up his age at the time.

In which year was Arthur born?

A 1909

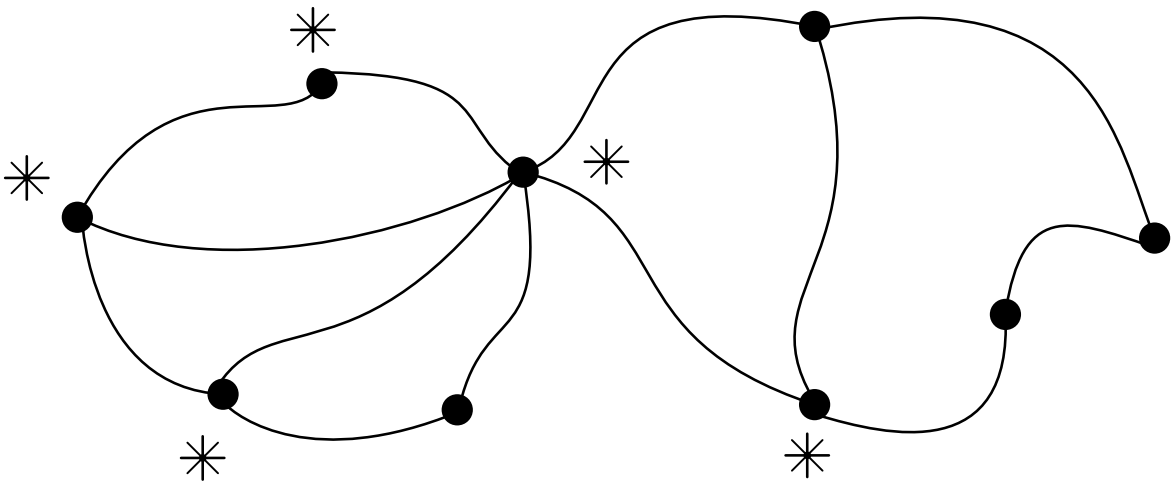
B 1916

C 1918

D 1936

E 1938

2 (2010 Q44) My local park has a network of paths linking various locations, as shown below. The stars on the map represent five of the park's landmarks: the fountain, lake, bandstand, statue and bowling green.



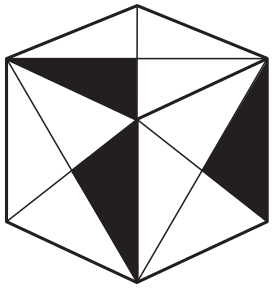
When I walk round the park, I often vary my routes and I do not always take the shortest route from one place to another. I have noticed that when I go from the statue to the fountain, I always pass the bandstand, but that going from the lake to the bowling green, I never pass the fountain unless I pass the bandstand twice.

How many of the stars on the diagram can be identified from the given information?

A 0
B 1
C 2
D 3
E 5

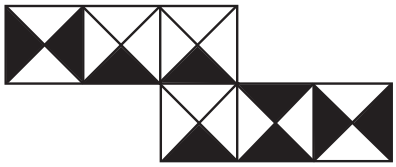
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3 Here is a diagram of a cube.
(2018 Q44)

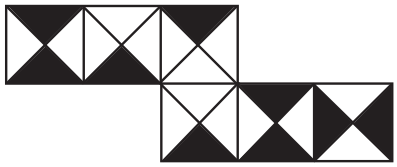


Which one of the following nets **could** be folded to make this cube?

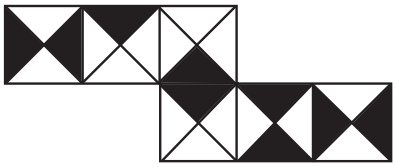
1



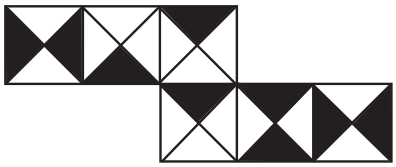
2



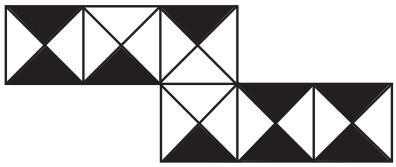
3



4



5



A 1

B 2

C 3

D 4

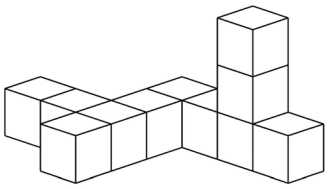
E 5

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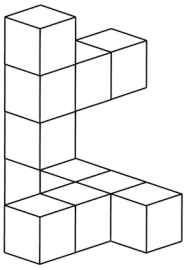
4 (2012 Q44) Five pictures of solids made by sticking cubes together are shown below.

Which one of the following (A-E) is a different shape from **all** the others?

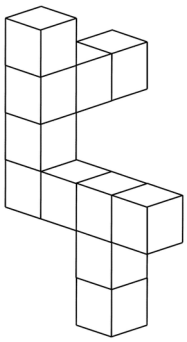
A



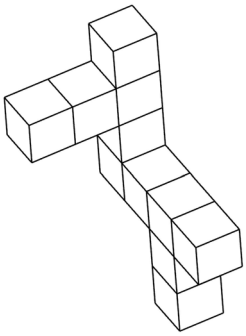
B



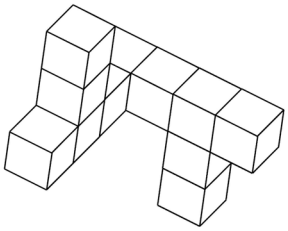
C



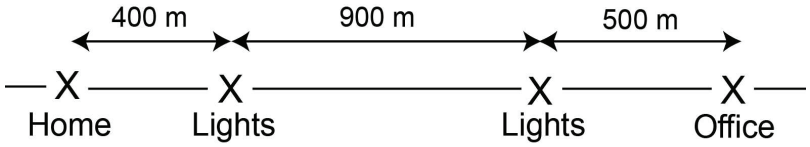
D



E



5 (2014 Q6) Central Avenue is a long straight road, several kilometres long. Both my home and my office are on Central Avenue, and the diagram below shows their positions, together with two sets of traffic lights that lie in between.



Both sets of lights turn to green for Central Avenue traffic simultaneously, showing green for exactly two minutes, turning green again exactly two minutes later.

I cycle to work every morning at a steady speed of 5 metres per second (except, of course, whilst waiting at traffic lights).

What is the **longest** journey time I can expect the journey to work to take?

A 6 minutes

B 7 minutes

C 8 minutes

D 9 minutes

E 10 minutes

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6 (2021 Q25) I run a bakery and sell two sizes of loaf. Small loaves (weighing 240 g) sell for 80p, and large loaves (weighing 400 g) sell for £1.20.

I buy dough, ready-made, at the start of each day. Small boxes (5 kg of dough) cost me £10.00 each, and large boxes (15 kg of dough) cost me £20.00 each.

The baking process causes each loaf to lose 20% of the weight of the unbaked dough.

I expect to sell 40 small loaves and 50 large loaves each day. I buy enough dough to be able to make these, and any unused dough at the end of the day has to be thrown away.

Assuming that I do not buy any more dough than necessary, what proportion of the dough that I buy each day do I have to throw away?

A 5.2%

B 6.3%

C 7.5%

D 17.8%

E 34.8%

7 (2020 Q32) Bert has agreed to make 14 decorations for his daughter's birthday. Each decoration will require two gold ribbons, one 12 cm long and the other 7 cm long.

The table shows the lengths and prices of available gold ribbon.

gold ribbon price list	
gold ribbon length	price
30 cm	£4.00
50 cm	£5.00
60 cm	£6.00

The ribbon can be easily cut to size for the decorations, but short pieces of ribbon cannot be joined to make longer lengths.

What is the least amount that Bert will need to spend on gold ribbon for the 14 decorations?

A £26.00

B £27.00

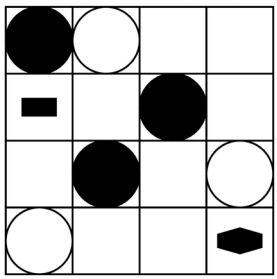
C £28.00

D £29.00

E £30.00

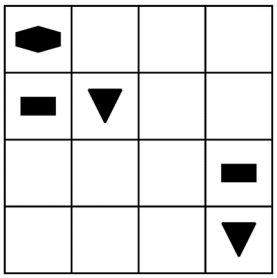
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8 (2012 Q14) In a game, circular counters are moved around a board which is a 4 x 4 grid. There are symbols on some of the squares on the board. At one point in the game, the board looks like this (all the circles represent counters which completely cover any symbol on the square that they occupy):

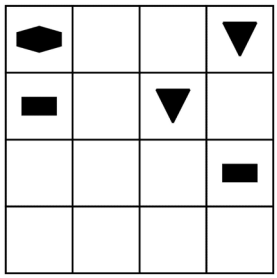


Which one of the following diagrams could show the layout of the board (not necessarily viewed in the same direction as above)?

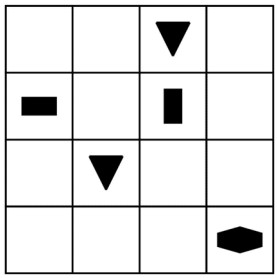
A



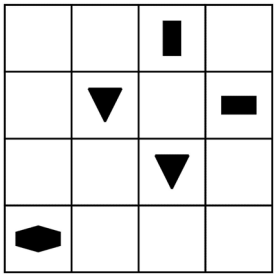
B



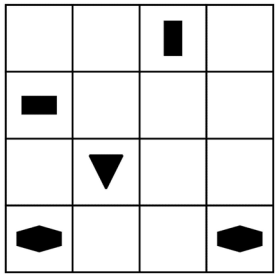
C



D

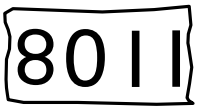


E



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9 (2010 Q49) All the telephone numbers in my village have six digits and begin with 58. I have accidentally torn up and thrown away a piece of paper on which I had written the telephone number of a friend who lives nearby. The only piece I can find is the scrap shown below:

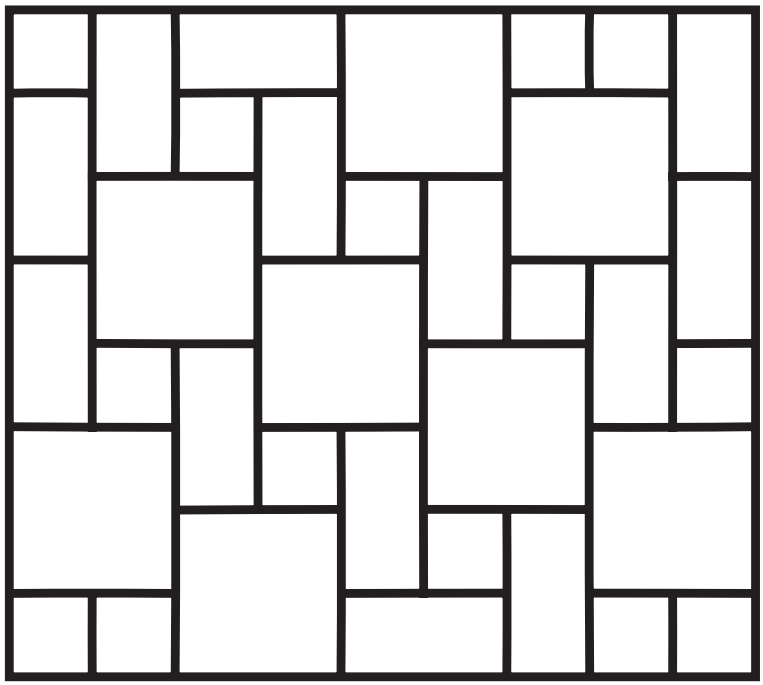


How many different phone numbers in my village could be consistent with this extract?

A 2
B 10
C 11
D 12
E 22

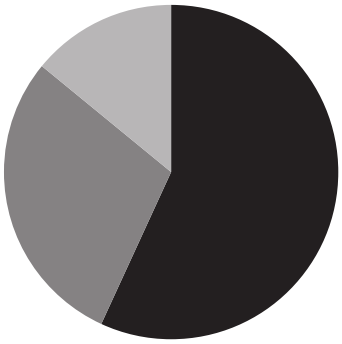
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10 The block paved area shown below is laid using three different-sized blocks.
(2023 Q20)



Which one of the following charts shows the relative comparison of the number of each size of block used in this paved area?

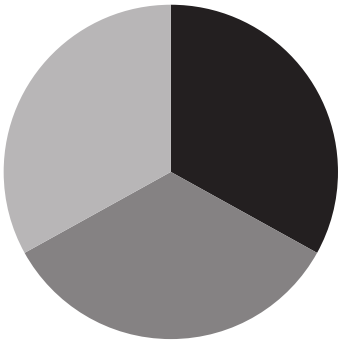
A



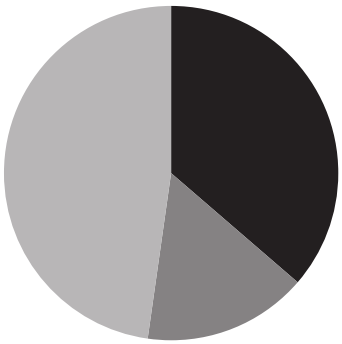
B



C



D



E



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11 (2017 Q36) Fergus Castle is a tourist attraction on a small island off the coast of the mainland. A ferry service runs between the mainland and the island, and takes 40 minutes to make a single journey in either direction. There is a turnaround time of 5 minutes at the end of each single journey. Each day, ferries leave the mainland every 20 minutes, at 5, 25 and 45 minutes past each hour, starting at 09:05. Ferries leave the island at 15, 35 and 55 minutes past each hour, starting at 09:15.

What is the minimum number of boats needed?

A 3

B 4

C 5

D 6

E 7

12 (2021 Q50) The company that I currently work for pays its employees monthly. Pay day is the fourth Thursday of every month or the 26th, whichever is earlier. Since I joined the company the dates on which I have been paid (in order) have been:

24th

24th

26th

26th

23rd

26th

25th (last month)

On what date will I be paid this month?

A 22nd

B 23rd

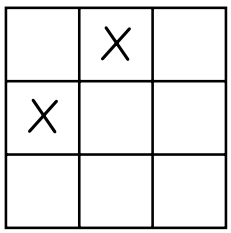
C 24th

D 25th

E 26th

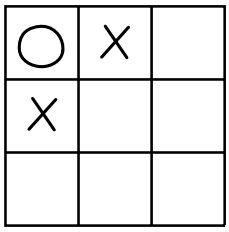
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13 (2011 Q31) In a modified form of a popular game, each of two players at any turn may put either 'O' or 'X' as she chooses in any blank square of a 3 by 3 grid. She wins if she thereby completes a line of three identical symbols. A player is faced with this situation:

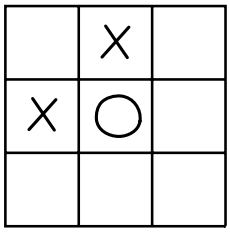


Which one of the following moves ensures that she can win at her next move?

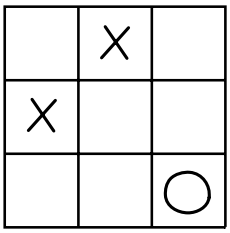
A



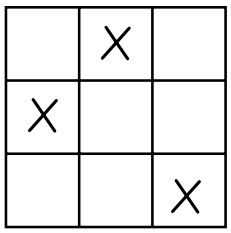
B



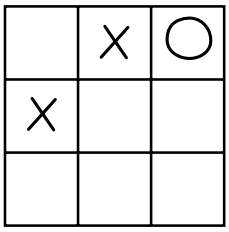
C



D

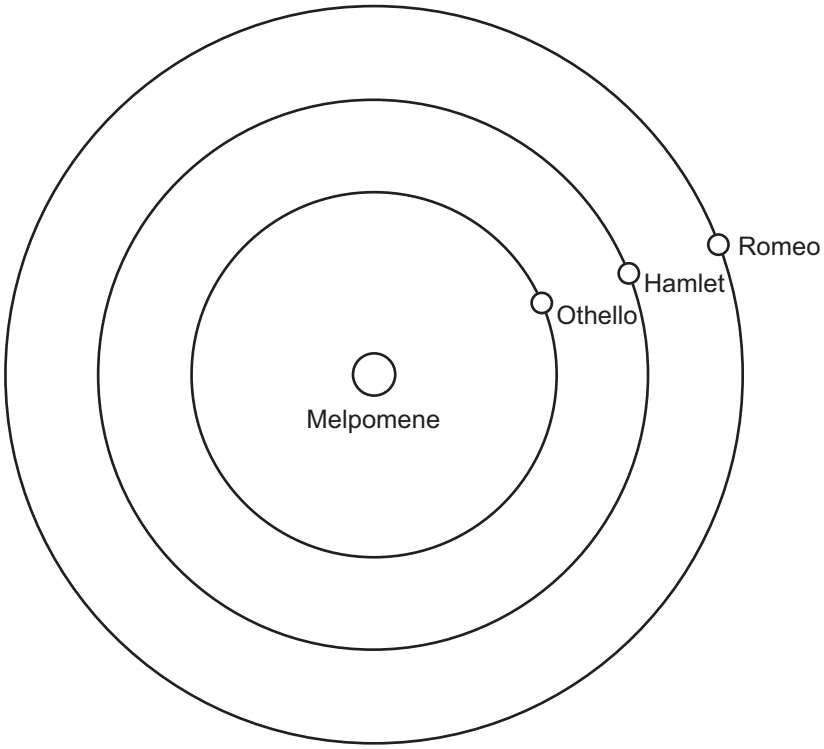


E



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14 (2016 Q48) The planet Melpomene has three moons, Othello, Hamlet and Romeo, each moving at a constant speed in the same direction in circular orbits.



Othello takes 20 days to complete one orbit of Melpomene, whereas Hamlet takes 45 days and Romeo takes 120 days.

How often are Melpomene and all three of its moons in line with each other in the order shown in the diagram?

A every 36 days

B every 72 days

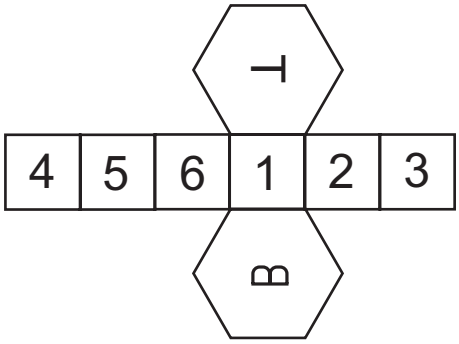
C every 132 days

D every 180 days

E every 216 days

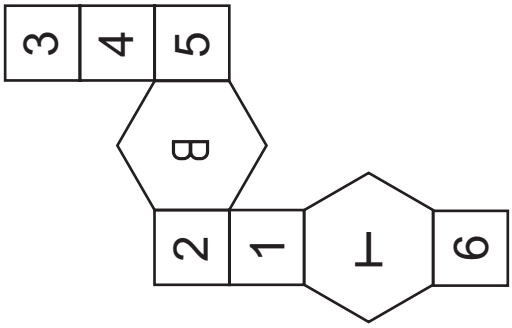
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15 Consider the following net, which forms a hexagonal prism:
(2019 Q44)

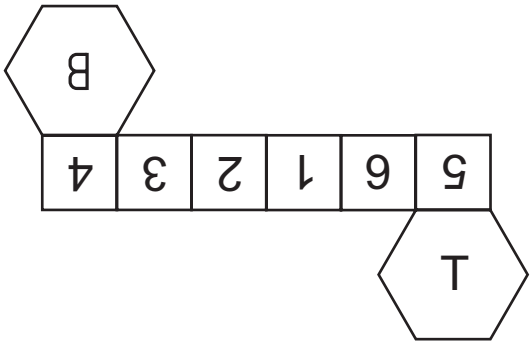


Which one of the following nets will form an identical prism?

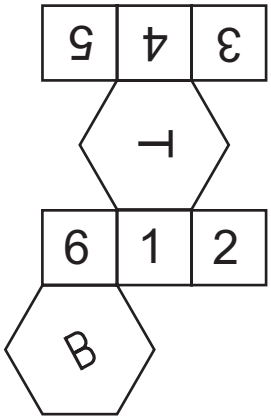
A



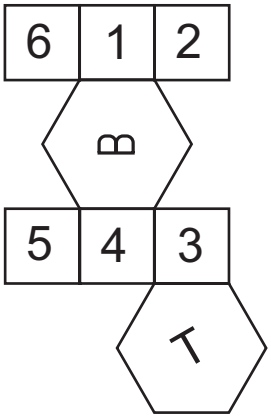
B



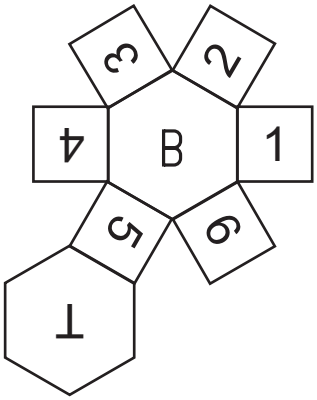
C



D

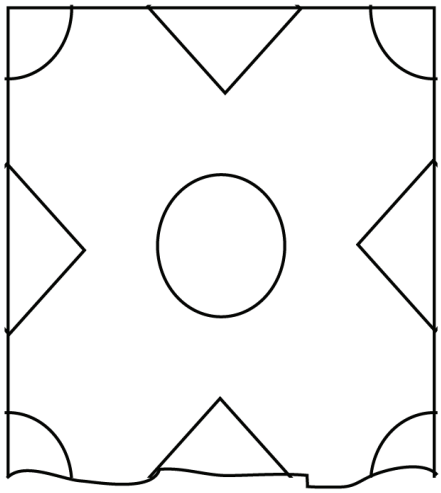


E



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16 (2016 Q50) Rolls of a particular wallpaper have a repeating pattern every 60 cm of their length. Each roll is 10 m long and 50 cm wide. A wall 2.5 m high and 6.2 m wide is to be papered from ceiling to floor with single lengths of paper hung vertically (drops).



How many rolls of wallpaper will be needed so that the pattern on each drop matches the next? (No joins can be made in one vertical drop.)

A 3

B 4

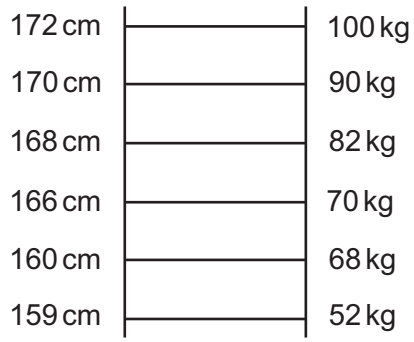
C 5

D 6

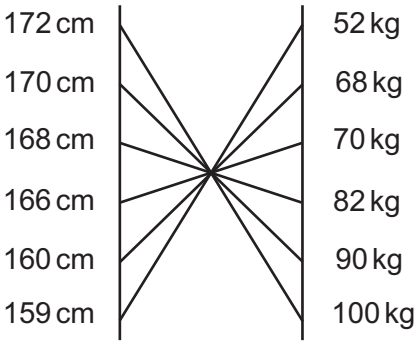
E 7

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17 (2018 Q50) Keith likes to draw ‘bump charts’ to look at relationships. For instance, if 6 individuals were ordered by height, and it happened that the order was the same by weight, the bump chart would look like this:

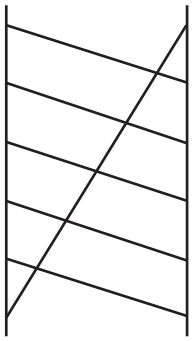


Or if height was maintained for order on the left hand but lightness rather than heaviness on the right hand, the bump chart would be this:

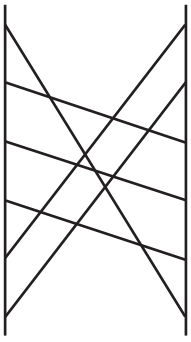


If one axis is reversed in order (as in heaviness goes to lightness), which one of the following bump charts would **not** be symmetrical about a vertical line midway between the axes?

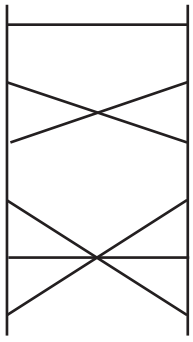
A



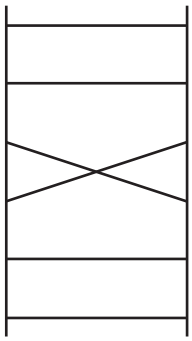
B



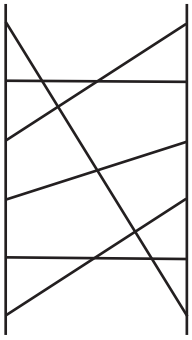
C



D



E



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18 (2018 Q30) Recently, packets of Amblers crisps have had 'money-off' coupons inside them. Some coupons are worth 9p, some are worth 14p and some are worth 20p.

George has been collecting these coupons. He has more 14p coupons than 9p coupons, and more 9p coupons than 20p coupons. The total value of all of his coupons is exactly £1.50.

How many coupons has George collected?

A 9

B 10

C 11

D 12

E 13

19 (2018 Q25) The President of Bugaria is elected using the supplementary vote system, which is conducted as follows:

Voters vote for their first-choice and second-choice candidates (though a second choice is optional).

The two candidates with the most first-choice votes go through to a second stage, whilst the others are eliminated. The second-choice votes for these two from the ballot papers of the eliminated candidates are added to their first-choice votes, and the candidate with the greater total is elected President.

The table below shows a summary of the votes cast for the five candidates in last week's Bugarian Presidential Election. For example, 37 862 voters cast their first vote for Sam and their second vote for Val.

	first-choice votes				
second-choice votes	**Sam**	**Val**	**Kim**	**Pat**	**Jo**
Sam	–	37 304	34 493	38 206	22 871
Val	37 862	–	25 517	27 818	20 143
Kim	23 148	31 690	–	30 247	42 620
Pat	26 493	24 485	29 681	–	31 354
Jo	28 506	27 530	25 308	24 763	–
no second choice	822	1027	1263	952	848

Who has been elected President of Bugaria?

A Sam

B Val

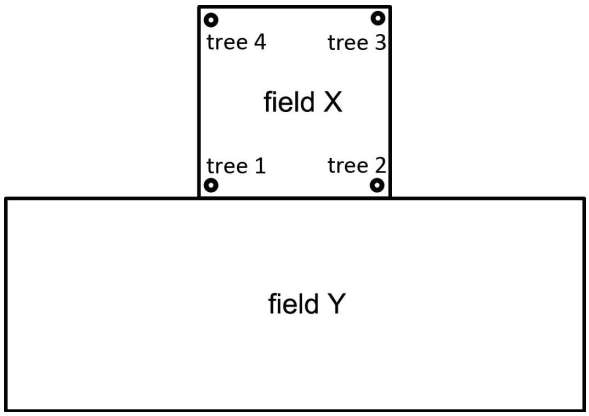
C Kim

D Pat

E Jo

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20 (2015 Q50) Four trees stand at the corners of square field X. They are photographed from field Y, from every possible direction, close up and from a distance, but never so that one tree stands in front of another.



In how many different orders, left to right, can the trees appear in the photographs?

A 1

B 2

C 3

D 4

E 5

21 (2020 Q46) My dog's coat had been looking dull, so I bought some conditioning tablets. The pet shop stocks these in boxes of 300 tablets, and for a dog the size of mine the recommendation (which I have followed) is 6 tablets per day.

Now, several months later, his coat is much shinier. Opening a new box of tablets today, I have decided to reduce his dosage. I shall just give him 6 tablets per day, twice a week, and on the remaining days each week he can have 4 tablets each day.

What is the maximum number of extra days that this box of conditioning tablets will now last?

A 10 days

B 15 days

C 16 days

D 24 days

E 25 days

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22 (2011 Q38) John wishes to have 500 copies of a brochure printed and is offered a choice of two price structures:

1. An equal cost for each brochure printed, regardless of number.
2. An initial fixed cost followed by a lower cost than structure 1 for each brochure.

Which one of the following sets of information would be sufficient for him to be able to decide which of the two price structures would be least expensive?

A The number of brochures for which the total bill would be the same by both structure 1 and structure 2.

B An example showing the cost of 500 brochures for structure 1 and 200 brochures using structure 2.

C The cost per brochure using structure 1 and the initial cost using structure 2.

D An example showing the cost of 200 brochures using structure 1 and 200 brochures using structure 2.

E The cost of 1 brochure using structure 1 and 2 brochures using structure 2.

23 (2013 Q50) The housing estate on which Andrew lives has streets of variable length. In each street the houses are directly opposite each other, with the same number of houses on each side. However, the numbering of the houses is inconsistent.

Some streets are numbered consecutively on each side, starting at one end of the street, going up to the end and then back down the other side.

Other streets have the odd numbers on one side, starting from 1, and the evens on the other side, starting with 2 at the same end.

Andrew's friend Amy lives on the same estate at 25 Acacia Avenue, which is not the end house in the street.

If Andrew knows the number of the house opposite Amy's, which one of the following statements is definitely true?

A He knows the exact number of houses in the street.

B He knows that the number of houses in the street is one of two numbers.

C If the house opposite Amy's is 26, he knows the number of houses in the street.

D If he knows the number of the house next door to Amy, he can work out the number of houses in the street.

E He knows the numbers of the houses immediately adjoining Amy's.

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24 (2014 Q37) My current membership of the squash club is about to expire and so I wish to renew it. The options available to me are given in the table below and have not changed since I joined as a new member six months ago (a change in membership type counts as a renewal, not a new member).

Membership type	Length	Extras	New member (£)	Renewal price (£)
Bronze	1 year	none	10	8
Silver	6 months	free locker	25	18
	1 year	free locker	45	28
Gold	6 months	free locker free competition entry	40	32
	1 year	free locker free competition entry	75	52

I am currently a Gold member, but I do not want to enter competitions any more as I play only once a month. I do need a locker when I visit and the normal cost to hire a locker is £2 per visit. I want to choose a membership that will give me the smallest possible total cost for a full year.

What will be the difference between the amount that I will pay now and the amount I paid six months ago?

A I will pay £32 less.

B I will pay £30 less.

C I will pay £12 less.

D I will pay £8 less.

E I will pay £5 more.

25 (2013 Q36) Last month, in an attempt to improve my fitness, I began to run daily. The maximum distance I have time to fit in on any particular day is 8 miles, but my aim is to maintain an average of at least 5 miles per day.

After 15 days my average was just under 6 miles per day, then the following day I increased my average to just over 6. Since then (for the last 9 days, up to and including today) I have been able to run only 2 miles each day.

What is the **minimum** number of days from now until I can bring my average back up to 5 miles per day?

A 2 days

B 4 days

C 9 days

D 18 days

E 26 days

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END OF TEST

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